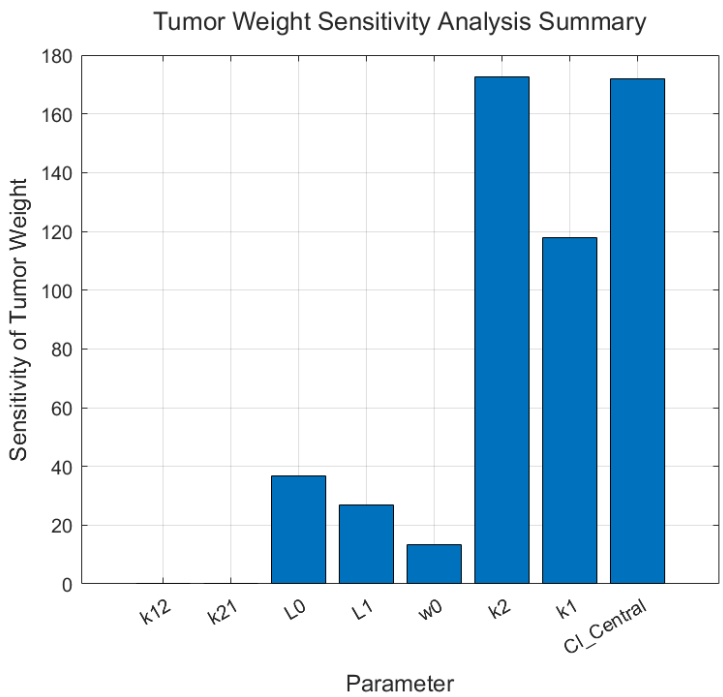
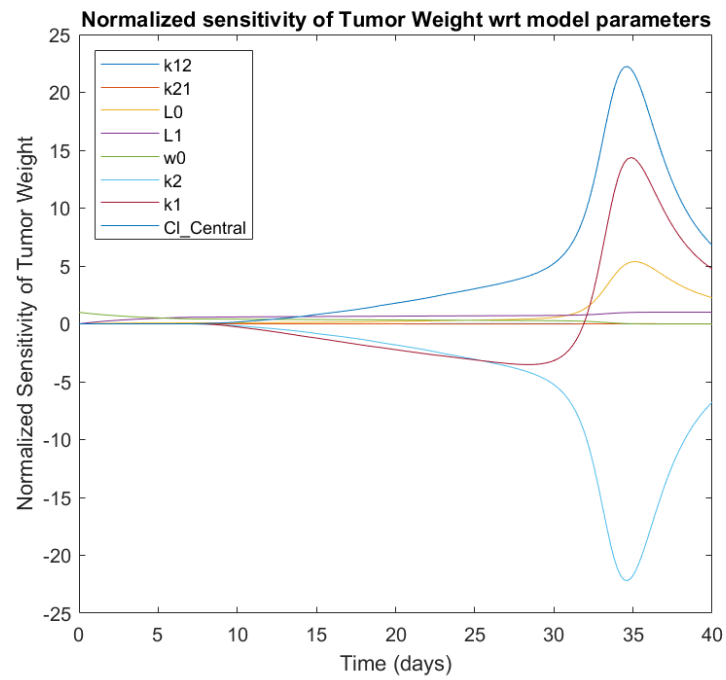
RBIF 108: Assignment 6

Jonathan Hummel
2025-05-13

Part 1

After following along with the MathWorks SimBiology tutorial on local sensitivity analysis using their example tumor growth model [1], I ended up with the plots below. The plot on the right was created by exporting the SimData object with the sensitivity analysis results to the MATLAB workspace and then generating a plot normally, using a snippet from another MathWorks tutorial as an example [2]:





As seen above, within this model, tumor growth is least sensitive to the parameters k_{12} and k_{21}, and most sensitive to the parameters k_2 and $Cl_{central}$. While the sensitivity of tumor weight to these parameters could be explored over a wider range of values, I did not do so because doing this for k_{12} and k_{21} in the tutorial led to the same conclusion. Below, I describe each of these least and most influential model parameters:

- ***k_{12}*** – This seems to be the parameter that the tumor weight is technically the least sensitive to (determined by zooming in on the rightmost plot, not shown). Looking at the tutorial and set of equations in the SimBiology Model Builder, this parameter is the rate constant for the transfer of the drug *from* the central compartment *to* the peripheral compartment, which is expressed as a linear function of the current drug concentration within the central compartment in the model equations. As such, k_{12} appears in Equations 1 and 2 of the system of ODEs that make up this model (see next page). The first equation describes the rate of change of the drug concentration in the central compartment, and the second describes the same for the peripheral compartment. Seeing as this system of ODEs is set up such that the drug in the peripheral compartment cannot act upon the tumor, one might expect this parameter to affect tumor weight more strongly (as well as k_{21}, discussed next). The fact that it does not could imply several things. For example, it likely means that, within this model, the weight/mass of proliferating

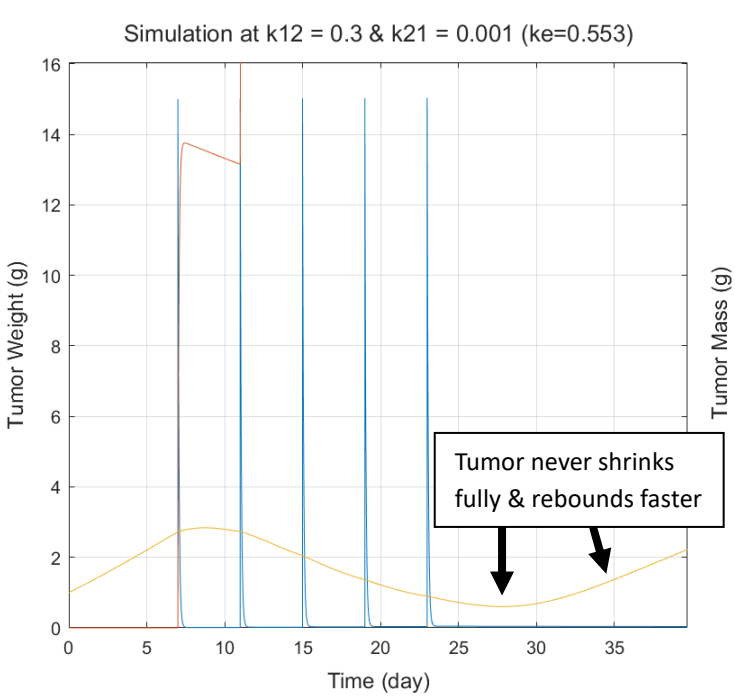
cancer cells (x_1) is sensitive enough to the drug (i.e. k_2) such that the initial spike in drug concentration from the dose delivered to the central compartment acts rapidly relative to the time the system takes to reach the equilibrium in which less drug exists in the central compartment due to transfer into the peripheral one (at least for the general ranges/combinations of all parameters considered here). Additionally, the linear elimination constant (k_e), which by default is much greater than either k_{12} or k_{21}, likely has a dominating impact on drug availability (discussed later).

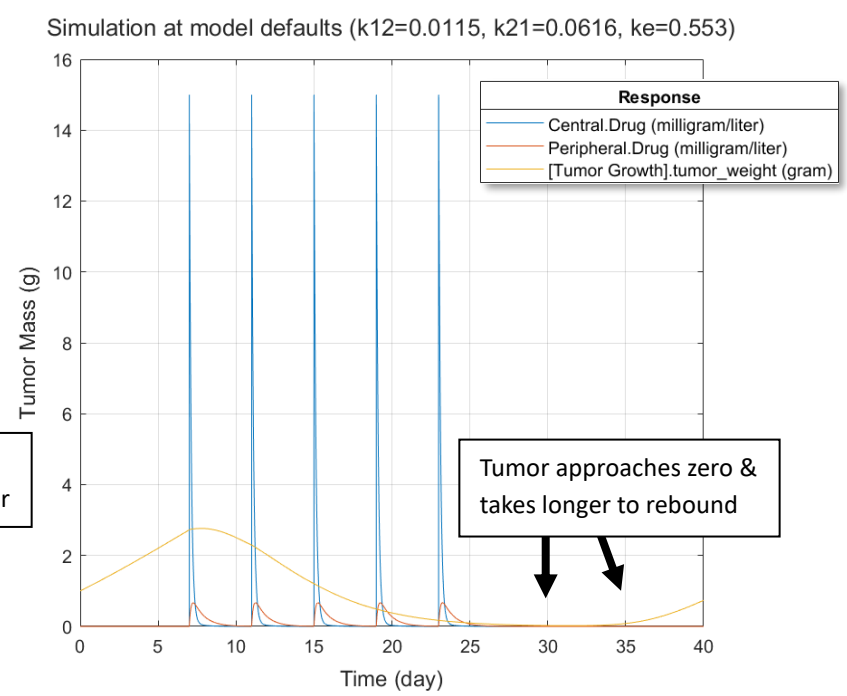
Equation 1: $$\frac{dD_{central}}{dt} = \frac{k_{21}D_{peripheral}V_{peripheral} - k_e D_{central}V_{central} - k_{12}D_{central}V_{central}}{V_{central}}$$

Equation 2: $$\frac{dD_{peripheral}}{dt} = \frac{k_{12}D_{central}V_{central} - k_{21}D_{peripheral}V_{peripheral}}{V_{peripheral}}$$

$D \equiv compartment\ drug\ concentration$
$V \equiv compartment\ volume$

- ***k_{21}*** – This is the parameter that the tumor weight is technically the second least sensitive to, but the sensitivity to this parameter is comparable to that of k_{12}. This parameter is the rate constant for the transfer of drug *from* the peripheral compartment *to* the central compartment, which can be seen above in Equations 1 and 2 from the model. Its default value is ~6x greater than the rate constant for the transfer of drug out of the central compartment (0.0616 hour^{-1} versus k_{12}=0.0115 hour^{-1}), meaning drug leaves the central compartment a bit slower than it enters by default. As noted above, k_e (a function of $Cl_{central}$) is much greater than either of these constants by default (~9x and ~50x greater), and so it likely has a more dominant effect than either of these on drug availability. That said, during the extended sensitivity analysis, there was minimal sensitivity even for k_{12} or k_{21} equal to 1 (~2x greater than k_e). After playing around with the model a bit (see below), it seems there are values of k_{12} relative to k_{21} and k_e for which the drug leaves the central compartment at a rate such that it becomes unavailable rapidly enough to matter, i.e., the minimum tumor weight reached after each treatment is noticeably farther from zero. That said, the ratio of k_{12} to k_{21} had to be pretty high (~300), and k_{12} had to be roughly comparable in magnitude to k_e for this to happen, which is consistent with the low level of sensitivity shown in the charts above. This observation is shown in the plots below (all other parameters are set to defaults):





On the left, the drug accumulates in the peripheral compartment, and the minimum tumor mass is affected.

- ***k_2*** – According to the sensitivity analysis, the tumor weight was most sensitive to this parameter. This makes sense, as it is the rate constant for the process by which drug in the central compartment sends proliferating cells (with total mass x_1) into the first non-proliferative state (with total mass x_2), which can not only hinder tumor growth by reducing x_1, but is also the start of the pathway toward death/removal of cells in this model (i.e., x_4 to sink). So, this parameter essentially dictates the drug's efficacy[1], and it comes as no surprise that it is the parameter that tumor mass is most sensitive to (see equations below).

  Equation 3: $$\frac{dx_1}{dt} = \frac{2L_1L_0{x_1}^2}{(L_1+2L_0x_1)\cdot x_{tumor}} - k_2D_{central}x_1$$

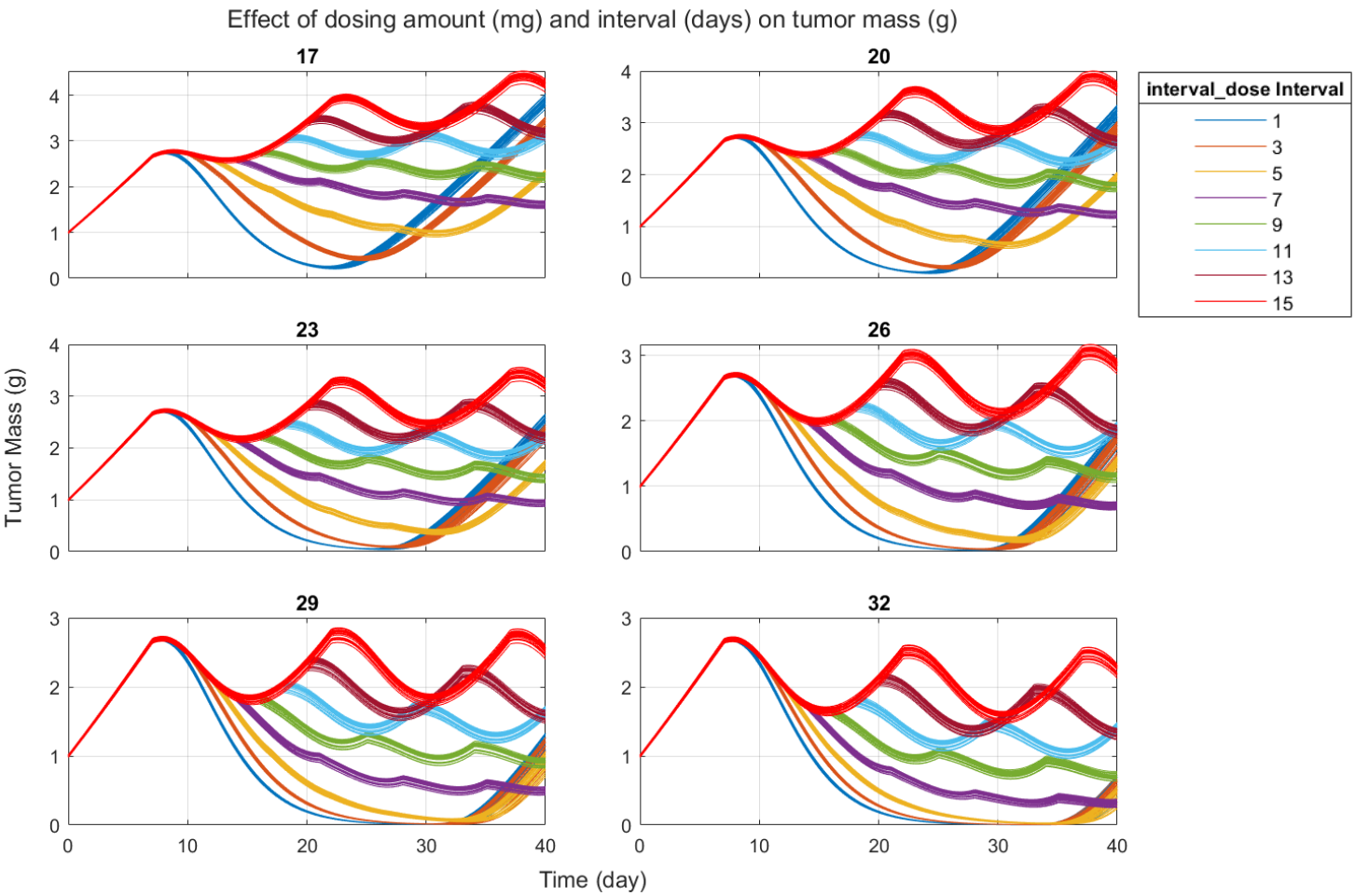
  Equation 4: $$\frac{dx_2}{dt} = k_2D_{central}x_1 - k_1x_2$$

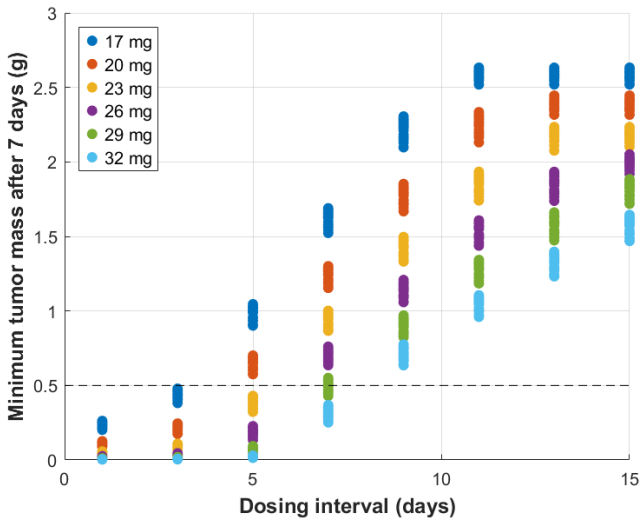
  $x_i \equiv mass\ of\ tumor\ fraction$, where i=1 is proliferating and i=2 is the first non-proliferating stage.

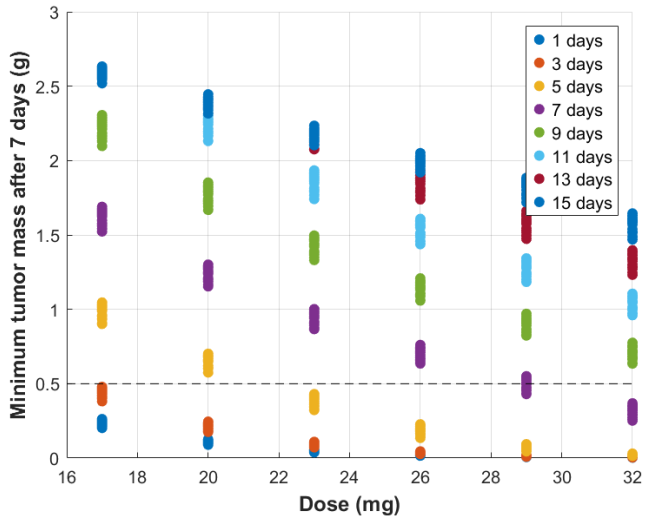
- ***$Cl_{central}$*** – as noted above, this was technically the parameter with the second highest sensitivity, but the impact was quite comparable to k_2 on the plots of sensitivity shown earlier. This also makes sense because, as mentioned earlier, the rate constant describing the elimination of the drug from the tumor (k_e) is a function of this clearance factor, $Cl_{central}$ [1]. With constant volume in the central compartment, k_e is proportional to $Cl_{central}$, and the role of k_e in the ODE system can be seen in Equation 1 (shown earlier). As discussed with regard to k_{12} and k_{21}, one might expect how long the drug persists where it has an impact (i.e., the central compartment) to matter quite a lot for the ability of the drug to affect tumor growth/mass. However, unlike the reversible process described by k_{12} and k_{21}, once the drug is cleared by the reaction controlled by $Cl_{central}$, it is gone for good from the system. This likely explains the very high sensitivity of tumor weight to this parameter compared to the relatively negligible sensitivity to k_{12} (or k_{21}) discussed earlier. In summary, a drug can be potent, but if it does not persist sufficiently long in the tumor, it will not be nearly as effective (and similar logic can be followed in the opposite direction). In other words, the residence time of the drug in the tumor can be just as important as its potency.

Part 2

After following along with the second tutorial, I obtained a plot showing the tumor responses for a simulated population (represented by n=25 stochastically-selected sets of model parameters) for a range of drug dose amounts from 5 to 35 mg (incremented by 6 mg) and a dosing interval of 4 days (the default). I also had a plot of the minimum tumor mass reached after 7 days (i.e., "stat1" from the tutorial) [3]. To explore the effect of varying both dosing amount *and* interval at the same time, I added the dosing interval variable as an additional component to the parameter set. I used dosing intervals from 1 to 15 days (incremented by 2 days). Also, I adjusted the range of dose amount to be 17 to 32 mg (incremented by 3) to focus on the optimal range from the tutorial in a little bit more detail [3]. Between the 25 patients, 6 dose amounts, and 8 dose intervals, this resulted in a total of 1200 scenarios simulated. The results from these simulations can be seen on the next page. The first set of plots shows the responses of tumor mass for the simulated patient population over time. Individual patient simulations are colored by the dosing interval, and each plot represents a different dose amount (subtitles are mg). The second figure is the same one obtained at the end of the optimization tutorial of minimum tumor mass after 7 days (i.e., "stat1"), but for combinations of varied dose amount *and* interval [3]. To make that second set of plots, I exported the samples and results from SimBiology into MATLAB and plotted them via a script.



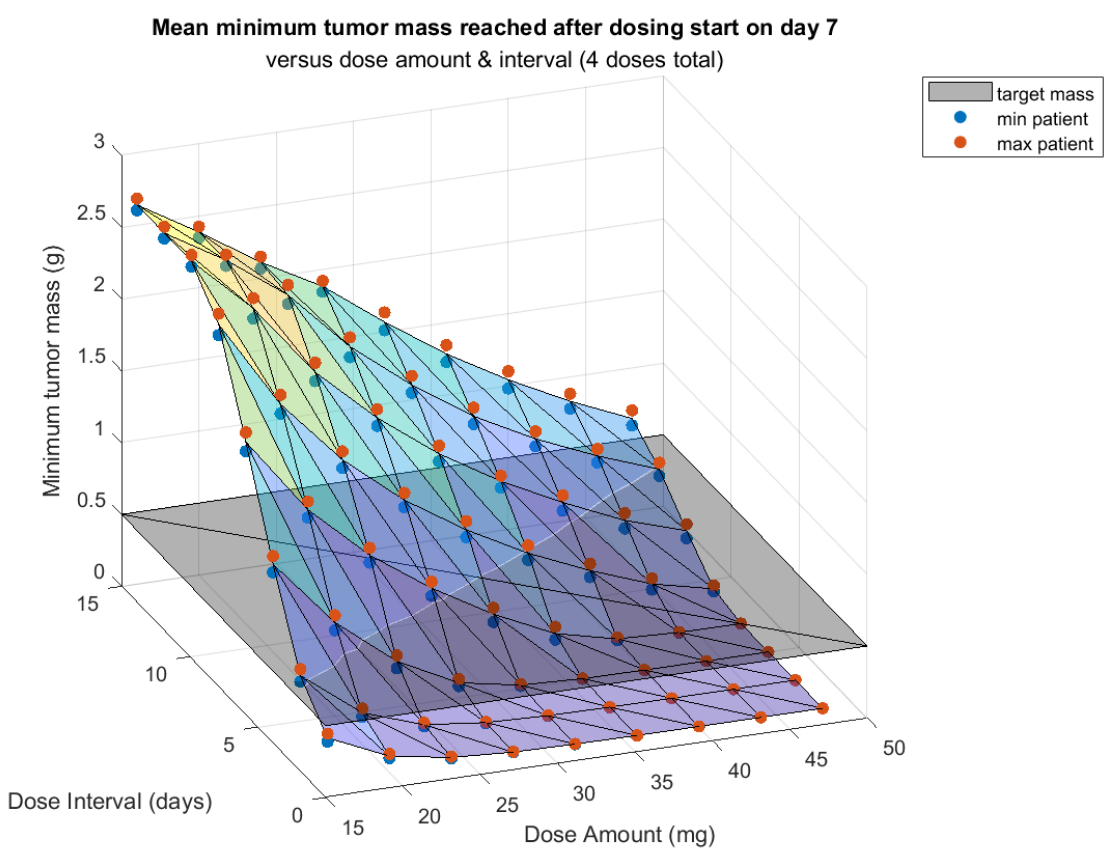




Because this is a hypothetical scenario, I will use the target tumor weight of 0.5 g from the tutorial as a reference point when discussing optimums (indicated by the dashed line on the bottom two plots). Somewhat unsurprisingly, the plots above point toward an optimum range at the highest dose amounts *and* shortest intervals between doses. It is worth noting that I did not change the total *number* of doses applied for these simulations, which was set to 4 by default in this model. Based on the bottom two plots, it looks like doses of roughly ≥ 23 mg spread ≤ 5 days apart constitute the cutoff to achieve a minimum tumor mass of ≤ 0.5 g after dosing begins on day 7 (first of 4 doses). This is not a strict rule. For example, it may be possible to use the highest dose simulated (32 mg) with a greater dosing interval (i.e., 7 days). Similarly, the lowest dose simulated (17 mg) achieves the objective when a 1-day interval is used. That said, the 6 plots of tumor mass over time show that simply reaching 0.5 g or less does not necessarily capture the full picture. This is where the solutions for tumor

mass over time become useful. For a dose of 17 mg and an interval of 1 or 3 days (top left), we can see that the tumor mass dips below 0.5 g, between days 20 and 30, then makes a fairly vigorous rebound later on, compared to 32 mg and the same dosing intervals (bottom right). In the latter scenarios, the tumor reaches a minimum mass much closer to 0, which is the ideal objective because we want it to die completely and not come back. In this continuous/idealized model, the tumor always comes back because it can only ever *approach* zero, but in reality, the cells that make up a tumor are discrete, and so there is a point where the tumor can be completely eliminated. So, perhaps the ideal dosing strategy is roughly ≥ 30 mg and ≤ 3 days between doses (≤ 5 days at most). The original model was already pretty much set within this range, as the dose was 30 mg and the interval was 4 days by default. That said, this type of optimization needs one important bit of context. In some ways, a dosing optimization analysis that merely suggests higher doses applied very often is not too helpful, because, of course, a tumor can be killed this way, however, patient safety is the ultimate constraint. Depending on the specifics of the treatment and the patient, ≥ 32 mg injections spaced only a few days apart could be quite tolerable, or it could be deadly or prohibitively harmful. Striking the balance between efficacy and patient safety is non-trivial for chemotherapy drugs [4-6]. The plots on the previous page represent a multidimensional surface (see plot below), and the optimal region of that surface would ideally be bounded by another surface representing the safest maximum exposure levels for patients as it translates to combinations of dose amount and interval, in addition to a target range of minimum tumor mass. In practice, such a surface would likely be more of a cloud/distribution. I did not vary the number of doses here (4), but that is yet another dimension that would probably need to be varied as well, within the constraints of patient safety, in order to find an optimal and feasible strategy.

To summarize, if we take a simplistic approach to the results above, and assume a target minimum tumor mass of ≤ 0.5 g using 4 doses, then doses of roughly ≥ 23 mg spread ≤ 5 days apart could achieve this goal. Within that range, higher dose amounts toward 32 mg (or greater) *and* lower intervals toward 1 day would likely improve the odds of complete removal. That said, how far we could go in that optimal direction would, in reality, be dictated by patient safety. Below is an extended optimization surface of the mean minimum tumor mass reached after the first dose, for a broader range of simulated dose amounts (plotted in MATLAB):



References

In addition to the resources cited in the Week 6 Lecture Notes by Professor Gongol:

[1] https://www.mathworks.com/help/simbio/ug/perform-sensitivity-analysis-for-tumor-growth.html

[2] https://www.mathworks.com/help/simbio/ref/simbiology.sensitivityanalysisoptions.html

[3] https://www.mathworks.com/help/simbio/ug/generate-virtual-population.html

[4] Morandi, P., Ruffini, P. A., Benvenuto, G. M., Raimondi, R., & Fosser, V. (2004). Cardiac toxicity of high-dose chemotherapy. Bone Marrow Transplantation, 35(4), 323–334. https://doi.org/10.1038/sj.bmt.1704763

[5] Extermann, M., Boler, I., Reich, R. R., Lyman, G. H., Brown, R. H., DeFelice, J., Levine, R. M., Lubiner, E. T., Reyes, P., Schreiber, F. J., III, & Balducci, L. (2011). Predicting the risk of chemotherapy toxicity in older patients: The Chemotherapy Risk Assessment Scale for High-Age Patients (CRASH) score. Cancer, 118(13), 3377–3386. https://doi.org/10.1002/cncr.26646

[6] van den Boogaard, W. M. C., Komninos, D. S. J., & Vermeij, W. P. (2022). Chemotherapy Side-Effects: Not All DNA Damage Is Equal. Cancers, 14(3), 627. https://doi.org/10.3390/cancers14030627